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Health and Human Services

HHS Health Sector AI RFI

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On behalf of the Association for perioperative Registered Nurses (AORN), we write in response to HHS's December 23rd request for information on what HHS can do to accelerate the adoption and use of artificial intelligence (AI) as part of clinical care.

As the professional membership association for perioperative registered nurses, AORN defines, supports, and advocates for patient and staff safety through exemplary practice in all phases of perioperative nursing care using evidence-based guidelines, continuing education and clinical practice resources. AORN's evidence-based *Guidelines for Perioperative Practice* address 36 perioperative practice areas and set the gold standard for interdisciplinary perioperative practice.

New and scheduled to be published May 21, 2026, is the AORN *Guideline for Integration of Artificial Intelligence*. Based on PRISMA systematic review methodology, this *Guideline* will first be available for public comment from Dec. 23, 2025 through January 31, 2026.

AORN agrees with the HHS on the importance of health care providers effectively using AI-enabled medical devices and tools to enhance productivity, reduce burden, lower health care costs, and improve health outcomes for patients, caregivers, and communities. We have adopted an evidence-informed position that is simultaneously optimistic and cautious. AORN believes that AI is suitable for augmentation of clinical decision making, but AI cannot and must never be a replacement for a human's clinical judgement. The human expertise, empathy and accountability required for clinical decision-making in practice cannot be replicated. Clinicians should be full partners in preparing for the future of AI that intersects their respective practice settings. AI integration methods matter – when integrated ethically and responsibly, AI has the potential to improve health care including patient outcomes, patient experience, and health care worker experience. When ineffectively integrated, AI has the potential to do harm. Risk mitigation should be prioritized in practice and risk mitigation strategies must be nimble and responsive to the ever-evolving AI risk landscape - both the known and unanticipated. AI education is essential and should be implemented in the academic setting and in the organizations where the AI-enabled technology is used. Competency should be assessed before using any AI-enabled technology.

Based on our *Guideline* authors' recent review of the literature and research during development of AORN's *Guideline for Integration of Artificial Intelligence*, AORN offers the following feedback:

Question 1: What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

Barriers for private sector innovation in AI adoption and use in clinical care exist in multiple forms:

- Lack of involvement of end-users (i.e., health care professionals whose job roles and responsibilities intersect with the AI tool used in clinical care) throughout the planning and development processes.

- Insufficient reporting and publication of AI tool development, testing, and validation processes.
- Limited ability for health care organizations to effectively evaluate, monitor, and maintain AI tools after deployment, including issues of interoperability with existing technology infrastructure and internal network capabilities.
- Regulatory uncertainty related to approvals and accountability.
- Resource constraints and issues of interoperability with existing technology infrastructure, especially in under-resourced health care organizations (e.g., rural, urban, non-academic medical centers).
- Clinician trust.

Each of these barriers contributes to the disconnect between the private sector enthusiasm, the influx of funding for novel AI-enabled technology, and real-world implementation in clinical care settings. Successful development, implementation, and maintenance of AI-enabled technology require early and intentional interdisciplinary collaboration among disciplines including technology developers and end-users (e.g., health care organizations, health care professionals) to identify and address internal and external barriers to innovative solutions that can enhance organizational efficiency, patient care delivery, and improve patient outcomes. Additionally, developers of AI-enabled technology must prioritize transparent reporting and publication of development processes, data collection and processing activities, and performance evaluation (i.e., internal and external validation of AI performance in real-world applications) to support user trust and adoption of AI in clinical care.

Resource constraints present a unique obstacle when integrating AI into clinical care settings. The literature indicates that many health care organizations are not currently equipped with technology infrastructure that is capable of meeting the increased demands for computational power and data-handling that AI-enabled technology requires.^{1,2}

Question 2: What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

HHS should prioritize changes to the Code of Federal Regulations related specifically to de-identification and re-identification of individually identifiable health information and the use of AI-enabled technology. Minimum necessary actions to limit and protect individually identifiable health information may vary according to the AI model type and architecture, individual health or other information being collected and used, and the amount of data required to re-identify an individual. To comply with § 45 CFR 164.514(b), additional considerations should be made to account for identifiers not already included in the current list of identifiers in § 45 CFR 164.514(b)(2)(i). With as few as four (4) pieces of data, a person may be reidentified, with that data not being exclusive to the items identified in paragraphs (b)(2)(i)(A) through (b)(2)(i)(R). Addressing this issue can provide health care organizations with additional guidance for data privacy and sharing practices that are required to protect individually identifiable health information and ultimately increase trust in the adoption and use of AI in clinical care.

Additional regulatory and policy solutions might include:

- requiring continuing education and professional development training in health care organizations specific to the safe use of the AI-enabled technologies deployed in the organization and used by health care workers;
- clarifying the requirements for “informed consent” when using AI assisted technologies;
- adopting a Total Product Lifecycle approach: iterative improvements while maintaining safety standards³;
- adopting Good Machine-Learning Practice: standards for ensuring quality and organizational excellence³;
- establishing requirements for safety reporting: a clearinghouse for adverse events similar to MedWatch for pharmaceuticals⁴;
- establishing requirements for standardized reporting of relevant AI model characteristics that explain the purpose, design, development, use, and limitations of the model similar to a nutrition facts label or model card; and
- establishing requirements for a Unique AI Identifier: a transparency label to enhance trust and accountability in AI systems.⁴

Question 3: For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

The deployment of AI in health care presents novel governance and accountability challenges, arising from unresolved questions regarding responsibility, liability, and oversight when automated systems fail or when risks go unrecognized. Current legal and regulatory frameworks do not yet provide clear guidance on how accountability should be assigned among developers, implementers, and end-users, particularly in situations where insufficient safeguards or unvalidated model behavior contribute to patient harm or degraded clinical performance (i.e., model drift). As a result, the absence of defined pathways for recourse or corrective action can impede both safety and trust. Addressing these gaps is essential to ensure that AI-enabled technology is integrated in a manner that supports consistent standards of care, transparent accountability, and responsible operational use.

Processes for implementation of AI-enabled technology are unique to the local context in which the technology will be deployed and require a robust internal governance framework to provide a structured, standardized approach for each phase of integration into the clinical setting (e.g., identification, evaluation, deployment, lifecycle management). HHS should dedicate resources to investigating the use and influence of the development and use of a governance framework in health care organizations on outcomes that are important to patients (e.g., quality of life, perceived health) and outcomes related to health conditions (e.g., surgical site infections, mortality, morbidity). HHS also should create mechanisms for mandatory reporting of serious adverse events linked to the use of AI-enabled technology in the clinical setting.

Question 4: For non-medical devices, what are the most promising AI evaluation methods (pre- and post deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms

would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

Several evaluation frameworks and reporting guidelines have been developed to facilitate health care organizations' evaluation of AI-enabled technology before widespread integration, including

- Model cards (i.e., standardized, plain-language document that explains the key characteristics of AI-enabled technology to understand what the model is, how it works, and how it should [and should not] be used),
- Expert/consensus guidelines for reporting and publishing AI-enabled technology development, testing, and validation processes (e.g., FUTURE-AI,⁵ CHEERS-AI,⁶ CONSORT-AI,⁷ CLAIM,⁶ MINIMAR,⁸ STARD-AI,⁶ SPIRIT-AI,⁷ TRIPOD+AI⁶), and
- Expert/consensus frameworks for health care professionals' evaluation of AI-enabled technology (e.g., PROCAST+AI,⁹ DECIDE-AI,¹⁰ CAIR Checklist,¹¹ 20 Critical Questions¹²).

Although they were developed by experts in relevant fields (e.g., data science, statistical analysis, computer science, use of the reporting guidelines, health care), the evaluation frameworks are yet to be validated for use and efficacy in positively influencing the evaluation and adoption of AI-enabled technology, as well as the influence of the use of these evaluation frameworks on patient outcomes and organizational outcomes (e.g., end-user satisfaction, workflow burden, usability, operational efficiency, financial impacts to the organization). HHS should dedicate resources to investigate the development and influence of the use of standardized evaluation frameworks for AI-enabled technology in health care organizations on these outcomes.

Question 5: How can HHS best support private sector activities (e.g., accreditation, certification, industry driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

HHS can best support private sector activities, effective and safe use of AI-enabled technology, and trustworthy AI by incentivizing the private sector to adopt activities and organizational priorities that promote trustworthy AI. The concept of trustworthy AI is widely published in the literature with much of the published literature referring to

- Transparency (i.e., extent to which individuals who interact with or use AI have access to information about the systems and their outputs),
- Reliability and validity (i.e., ability to consistently perform as required and intended, without failure),
- Accountability (i.e., responsibility of the individual(s) or organization(s) involved in the design, development, evaluation, deployment, and maintenance of AI-enabled technology to uphold safe, ethical, and equitable health care delivery practices),
- Fairness (i.e., ethical and equitable development and use of AI-enabled technology to prioritize that all patients receive accurate, unbiased, and high-quality care, regardless of background, identity, or circumstances),
- Utility (i.e., extent to which AI-enabled technology can be used to achieve specific goals and objectives to improve patient outcomes, clinical decision-making, and health care delivery), and

- Fitness of purpose (i.e., measure of how well AI-enabled technology meets the goals and objectives of the deploying organization based on the intended use, requirements and preferences, and organizational priorities).

Facilitating or requiring accreditation, certification, or credentialing of organizations that develop and distribute AI-enabled technology for use in the health care sector (herein referred to as ‘AI developers’) would incentivize these entities to adopt evidence-informed and ethically grounded practices for developing, testing, validating, and transparently reporting all relevant information about AI-enabled products (e.g., data collection and preprocessing practices, use of representative and diverse datasets, transparent performance and validation processes, benefits and limitations, intended and unintended uses, requirements for technical resources). By incentivizing AI developers to adopt an industry standard for accreditation or similar, health care organizations would be encouraged to seek out and prioritize AI developers that have demonstrated rigorous, sanctioned development processes, thereby enhancing trust and adoption of these innovative technologies.

Question 6: Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

Published literature detailing the real-world performance and cost of AI-enabled technology is limited, with much of the literature reporting feasibility studies and small, nonexperimental studies with limited generalizability. Additional research is required to investigate the large-scale adoption and effects of AI-enabled technology in clinical settings. Some examples of AI-enabled technology performance that has fallen short of expectations for efficacy, safety, and generalizability include the integration of a sepsis early warning system developed and integrated with an electronic health record platform,¹³ and AI-enabled technology to enhance detection and removal of colorectal polyps and adenomas during gastroenterology procedures.¹⁴

Artificial intelligence tools that would have the greatest potential to improve health care outcomes are tools that are able to provide outputs that are transparent and explainable by the end-user to highlight which input features most influenced a model’s prediction. Explainability is crucial for supporting trust in the AI model, regulatory oversight, bias detection, error identification and mitigation, and improved adoption by clinicians who need to understand the basis of AI outputs, as autonomous AI-enabled technology is not encouraged in patient care delivery.

Question 7: Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Primary influencers on the adoption of AI for clinical care are the intended end-users. Published organizational experience reports¹⁵ detail the influence and necessity for early inclusion of health care professionals whose roles and responsibilities intersect with AI use. Involvement and prioritization of the needs, preferences, and priorities of the end-user can improve the development process by including visual displays, alert type and frequency, and even technology platforms that incentivize the end-user to use AI-enabled technology as intended. This involvement also facilitates trust in technology by repeated

exposure to and use of the technology, explanation of the potential benefits and consequences of use, and the factors of use that matter most to the end-user (e.g., visual displays, ease of use, education, effects on user workflow). Other roles that influence adoption of AI include individual(s) in the organization with the responsibility and authority to oversee the evaluation, integration, and maintenance of AI-enabled technology. Organizational experience reports detail the creation of health care executive roles dedicated to these tasks (e.g. Chief Health AI Officer). This role can serve as the executive presence and authority to develop, maintain, and update an AI governance framework across the organization and should remain a separate and dedicated role.

Administrative hurdles to the adoption of AI in clinical care are numerous. Fragmented organizational processes for identifying, evaluating, and deploying AI-enabled technology present concerns for inadequate governance oversight at the organizational level, in addition to duplicative or inefficient evaluation or deployment processes. Lack of familiarity with relevant regulations, competing priorities and incentives for adoption across an organization, and lack of validated governance framework present additional challenges for health care organizations to overcome.

Question 8: Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Enhanced interoperability to accelerate the development and adoption of AI-enabled technology in the health care setting requires industry adoption for reporting of the different phases of the lifecycle of AI. Regulation for the use of standardized AI model cards (*see Response to Question 4 for additional information*) or other reporting requirements encourage transparent development processes and consumer evaluation prior to purchase and deployment. Additionally, development of large, de-identified national data sets for the use and development of AI-enabled technology may enhance interoperability and accelerate AI research. Federated learning as a strategy for anonymizing data and training AI models through decentralized data repositories can enhance accessibility to datasets and improve privacy.

Question 9: What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

AI has the promise to revolutionize the health care sector by rapidly increasing efficiency, decreasing administrative burden, and augmenting clinical decision-making to improve patient outcomes. Caregivers and patients alike believe that AI has the potential to offset administrative tasks with more time for genuine patient interaction and connection. This is coupled with concerns for privacy (e.g., ambient listening technology being used to identify or scrutinize patients or caregivers), self-determination, retaliatory or discriminatory care delivery based on protected characteristics, and a loss of the human element of health care delivery.

Question 10: Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

All research should prioritize improvement of patient outcomes and clinician user experience with other outcomes recognized as secondary (eg, cost savings, efficiency). In other words, the research should be

structured so that the aim is always to improve the human (ie, patient and clinician) experience above other perceived improvements. Areas of AI research the HHS should prioritize to accelerate the adoption of AI as part of clinical care include standardized methods for reporting and evaluating AI-enabled technology pre- and post-deployment, effects of implementing a structured governance framework on organizational operations and patient outcomes, methods for detecting and mitigating bias in the development and use of AI-enabled technology, implications for use of agentic AI tools in clinical care, and the use of AI to reveal factors contributing to adverse events in clinical care.

Question 10(a): Are there published findings about the impact of adopted AI tools and their use in clinical care?

A substantial and rapidly expanding body of published research and other reports examines the clinical impact of adopted AI-enabled technology across diverse care settings. Systematic reviews and health-economic evaluations report improvements in diagnostic accuracy, workflow efficiency, and patient outcomes, although the strength and quality of evidence vary across specialties and methodologies.^{16,17} Recent reviews also note that while many studies demonstrate positive effects, the evidence base remains heterogeneous and would benefit from more rigorous, real-world evaluations. Overall, the literature provides clear indications that AI tools already in use have measurable clinical and operational impacts, though continued research is needed to assess long-term performance and safety. The AORN Guideline for Integration of Artificial Intelligence includes a section dedicated to examples of clinical application of AI enabled technology that have been studied in the perioperative practice setting.¹⁸

Question 10(b): How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

Costs and benefits of AI must be approached on an individualized and ongoing basis; unique to each organization, governance structure, and AI-enabled technology being integrated into the organization. Costs and benefits are characterized by identifying improved diagnostic accuracy, reduced clinician burden, effects on patient outcomes, efficiency or time savings, and frequency of adverse events. There is a need for more comprehensive and methodologically consistent evaluations of the costs and benefits. Additionally, the literature indicates that AI adoption not only generates costs and benefits but also shifts resource allocation, labor distribution, and financial responsibilities across stakeholders, although these transfers are less consistently measured than direct costs or benefits.

Should you have any questions, please feel free to contact our Senior Director of Evidence-Based Perioperative Practice:

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Sincerely,



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